

Michael Beirne

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Objective

Interested in building reliable systems, translating large datasets into useful products, and applying computer science and mathematics to high-impact problems.

Education

Columbia University, Columbia College - New York, NY

May 2028

Degree: B.A. candidate in Computer Science and Mathematics

Academic profile: GPA: 3.50/4.00; SAT: 1530 (790 Mathematics)

Relevant study: Data Structures and Algorithms; Computer Systems and C Programming; Discrete Mathematics; Linear Algebra; Probability and Statistics; Multivariable Calculus; Modern Algebra; Machine Learning; database systems and software design through independent study and project work.

Organizations: BlackGen Capital, BLK Capital Management, Columbia EESA, Putnam Seminars, National Society of Black Engineers, Columbia Undergraduate Mathematical Modeling Society.

Honors: Dean's List; American Association of Blacks in Energy Scholarship; Ron Brown Scholar Program - Navigator Scholarship; Columbia Named Scholar.

Professional Experience

NASA L'SPACE Academy - New York, NY

Sep. 2026-Dec. 2026

Incoming Data Science Research Fellow

Selected for a fall research program focused on applying technical analysis, collaborative problem solving, and project execution to NASA-oriented scientific and engineering work.

Expected to contribute to a multidisciplinary research project through data analysis, written technical communication, milestone reviews, and a final presentation of findings.

SoFi - Remote

Aug. 2026

Extern

Selected for an August 2026 externship offering exposure to financial technology, product strategy, and the use of data to improve consumer financial services.

Engaging with professionals and applied business cases to deepen understanding of how software, analytics, and customer-focused product decisions intersect in fintech.

Enterall - New York, NY

Jul. 2026-Present

AI / ML Engineer Intern

Contributing to an AI-native creative workspace for fashion professionals, with emphasis on converting bitmap images into vector drawings with state-of-the-art precision.

Investigating computer-vision and generative-model workflows that preserve garment structure, contours, and visual detail while producing editable vector assets.

Working within an early-stage product environment where model quality, interface behavior, and user-facing creative workflows must be evaluated together.

The Riverside Company - New York, NY

Jun. 2026-Present

Private Equity Summer Analyst

Evaluate prospective investments through sector research, company analysis, competitive landscaping, and review of operating and financial performance.

Support leveraged buyout analysis by examining revenue growth, margins, cash generation, capital structure, debt paydown, and potential return outcomes.

Synthesize market information and diligence findings into concise materials used to discuss new opportunities and key investment risks.

BeReal - New York, NY

May 2026

Product Management Extern

Mapped user journeys and engagement flows to identify friction points affecting activation, repeat usage, and retention.

Designed and prototyped a retention-oriented feature intended to improve monthly active usage, refining the concept through benchmarking, wireframing, and iterative feedback.

Presented the product rationale, intended user behavior, and success metrics in a concise final recommendation.

Capital One - Auto Navigator - McLean, VA

Jun. 2025-Aug. 2025

Product & Analytics Intern

Developed recommendation logic using behavioral embeddings, ranking heuristics, and interaction analytics to improve how users discover relevant automotive content and offers.

Analyzed page visits, event tags, and behavioral patterns to identify signals that could strengthen personalization and recommendation quality.

Collaborated with backend engineers and product partners to prototype features, define user interactions, and iterate based on technical constraints and observed behavior.

Translated analytical findings into product recommendations for stakeholders, balancing customer value, implementation feasibility, and measurable outcomes.

Capital One - Commercial Real Estate M&A - McLean, VA

Jun. 2024-Aug. 2024

Business Analyst Intern

Used proprietary credit-card transaction data spanning more than 100 million transactions to model and forecast macroeconomic trends relevant to commercial real estate and banking.

Cleaned, aggregated, and interpreted large datasets to identify changes in consumer behavior, sector activity, and geographic performance.

Built an analytical narrative connecting transaction-level patterns to broader market conditions and presented the results to stakeholders as actionable business insight.

Jane Street - New York, NY

Jun. 2023-Aug. 2023

Summer Fellow - Academy of Mathematics and Programming

Completed an intensive program at the intersection of mathematics, computer science, and financial markets.

Studied combinatorics, number theory, programming, probability, and market-making through lectures, problem sets, and trading simulations.

Developed comfort solving unfamiliar quantitative problems under time constraints and communicating reasoning clearly with peers.

Technical Projects

Research Terminal Platform

Independent Project

Built a quantitative research platform for exploring market data and estimating ordinary least squares factor-exposure models.

Designed a layered caching architecture with time-to-live in-memory storage and an on-disk cache to reduce repeated external requests and improve reliability.

Structured the system around reusable data-access, modeling, and presentation components so new assets, factors, and analyses can be added without rewriting the entire application.

Focused on practical engineering concerns including cache invalidation, persistence, API failure handling, reproducible calculations, and clear output for research workflows.

NAND to Tetris Computer System

Independent Project

Built a complete 16-bit computer system beginning with NAND gates and progressing through a working hardware and software stack.

Implemented HDL logic chips, arithmetic components, memory, a CPU, and the full computer architecture, then wrote an assembler and virtual-machine translator.

Implemented a compiler for the Jack programming language and core operating-system libraries, connecting high-level programs to the underlying hardware platform.

Used the project to develop a systems-level understanding of abstraction layers, instruction execution, memory, compilation, and operating-system services.

FindJobs - Internship Aggregation Website

Independent Project

Built and deployed a full-stack internship aggregation platform using JavaScript, HTML, CSS, and GitHub Pages.

Organized listings into a searchable interface intended to reduce the time students spend locating relevant opportunities across fragmented sources.

Handled front-end presentation, data organization, deployment, and iterative improvements to usability and information clarity.

Backend and Distributed-Systems Practice Portfolio

Independent Projects

Developed Java and Spring-based applications including a JWT-secured to-do API, caching proxy, weather API with Redis, unit-conversion service, and GitHub activity command-line tool.

Practiced layered architecture, dependency injection, REST API design, authentication, database integration, Docker-based deployment, and error handling.

Built a WebSocket broadcast server and Docker Compose applications to strengthen understanding of networking, concurrency, service communication, and containerized development.

Research and Technical Programs

Project Hedgehog - Evaluations - Remote

2026-Present

AI Researcher

Evaluate outputs from multiple AI models to improve instruction following, formatting, correctness, and overall response quality.

Apply detailed rubrics, compare candidate responses, identify failure patterns, and provide structured judgments that support model improvement.

Developed stronger habits for verifying AI-generated work rather than accepting confident but unsupported outputs.

Google Developer Program - Remote

2026

Developer Participant

Completed codelabs and practical exercises involving Google Cloud, Firebase, APIs, and machine-learning tools

Built small projects to understand how cloud services, data storage, authentication, and application interfaces fit into deployable products

Leadership, Finance, and Campus Activities

Major League Hacking (MLH) Fellowship	2026-Present
Fellow	
Selected for a professional-development fellowship providing structured coaching, career preparation, and access to a nationwide network of peers and industry partners.	
Participate in skill-building, interview preparation, mentorship, and leadership programming designed to support long-term professional growth.	
BlackGen Capital and BLK Capital Management - Columbia University	2024-Present
Member	
Participate in finance education and investing activities focused on company research, valuation, markets, and professional development.	
Contribute to a community supporting underrepresented students pursuing careers in investing, banking, private equity, and related fields.	
Columbia Undergraduate Mathematical Modeling Society - New York, NY	2024-Present
Member	
Participate in discussions and exercises on mathematical modeling, estimation, and applications of quantitative methods across scientific and real-world problems.	
National Society of Black Engineers - Columbia University	2024-Present
Member	
Engage in technical, academic, and career-development programming with a community of engineering and computer-science students.	
Putnam Seminars - Columbia University	2024-Present
Participant	
Practice proof-based problem solving in algebra, combinatorics, number theory, and analysis through challenging competition-style mathematics problems.	

Memberships

Ron Brown Scholar Program
Telluride Association
SEO Scholars
Management Leadership for Tomorrow (MLT)

Technical Skills

Programming languages: Java, C, Python, JavaScript, TypeScript, C++, SQL, HTML, CSS, JSON, Assembly.
Frameworks and tools: Git, Linux, Docker, Kubernetes, React, Bash, AWS EC2, PostgreSQL, Redis, Spring Boot, Node.js, Figma, Jupyter.
Technical areas: Data structures and algorithms, backend development, REST APIs, databases, caching, systems programming, machine learning, computer vision, product analytics, quantitative modeling.

Awards and Additional Information

Dean's List, Columbia University.
American Association of Blacks in Energy Scholarship.
Ron Brown Scholar Program - Navigator Scholarship.
Columbia Named Scholar.
Interests include tennis, chess, reading, cooking, exploring new cuisines, volunteering, and Citi Bike rides throughout New York City.

Selected Additional Projects and Independent Study

JWT-Secured To-Do API: Built a Java and Spring Boot REST API with authentication and authorization, persistent task storage, structured request validation, and clear error responses. Used the project to practice controller-service-repository separation and secure backend design.

Weather API with Redis Caching: Created a service that retrieves weather data from an external provider, caches responses in Redis, and returns normalized results through a simple API. Considered expiration policy, failed upstream requests, and reducing unnecessary network calls.

Caching Proxy Server: Implemented a proxy that forwards requests to an origin server and stores reusable responses, first in Node.js and later as a Java/Spring rewrite. Explored cache keys, time-to-live behavior, command-line configuration, and transparent request forwarding.

WebSocket Broadcast Server: Built a command-line broadcast server supporting multiple connected clients and real-time message distribution. Used the project to study sockets, connection lifecycle, concurrent clients, and server-side coordination.

Docker Compose Application: Containerized an application and its database as separate services, configuring networking, environment variables, persistent storage, and repeatable local startup through Docker Compose.

Machine Learning Study: Worked through an end-to-end housing-price modeling workflow involving data loading, stratified train-test splits, exploratory analysis, missing-value imputation, categorical encoding, feature scaling, pipelines, model evaluation, and cross-validation in Python and Jupyter.